

Low Level Design (LLD)

Playwright BDD Cucumber Framework · Version 1.2 · Updated 16 Aug 2026

1. Directory map

src/config	env, artifacts modes, browsers, BrowserStack, run IDs, paths
src/world	CustomWorld (page, actions, business getters, shared, stepIndex)
src/hooks	Before/After/AfterStep lifecycle, artifacts, triage
src/business	BaseBusiness, LoginBusiness, CustomerBusiness, ApplicationBusiness
src/pages	BasePage, LoginPage, AddCustomerPage, CustomersPage, OpenAccountPage, CustomerDashboardPage, templates/
src/utils	PlaywrightActions, WaitConditions, Assertions, SelfHeal, waits, files, dates, secretCrypto
src/auth	Optional manager session helpers (storageState)
src/errors	FrameworkError hierarchy + mapper
src/logger	Winston factory
src/shared	SharedBuffer (scenario/worker/global)
src/db	MySQL/Postgres adapter, schema, repository
src/reports	Custom HTML, JSON history, extent/ activity HTML
src/ai	llmClient, modelRouter, FailureTriage, runTriage, llmHealth
features/	Gherkin + step definitions (login, customer)
recordings/	recorded executions/ (codegen sources), test-runs/ (scenario videos)
dashboard/	Express API + Chart.js UI
docker/	Dockerfile and compose
.github/	GitHub Actions
jenkins/	Jenkinsfile
docs/	HLD/LLD HTML and PDF

2. Configuration contract

Loaded from `.env` via `src/config/env.ts`.

Area	Keys
App	BASE_URL, APP_USERNAME, APP_PASSWORD, APP_ENCRYPTION_KEY
Browser	BROWSER, BROWSER_CHANNEL, HEADLESS, KEEP_BROWSER_OPEN, SLOW_MO, PARALLEL, LOGIN_PAUSE_MS, CLICK_PAUSE_MS
Artifacts	SCREENSHOT, STEP_SCREENSHOT, SCREENSHOT_FULL_PAGE, VIDEO, TRACE — each off on retain-on-failure

Self-heal	SELF_HEAL_ENABLED (alias SELF_HEAL), SELF_HEAL_USE_LLM, SELF_HEAL_TIMEOUT_MS
LLM	LLM_ENABLED is the on/off switch. OLLAMA_ENABLED is the same flag (alias), not a second setting. Host/model/key: LLM_HOST, LLM_API, LLM_API_KEY, LLM_MODEL, plus LLM_ROUTE, LLM_ROUTER_MODEL, LLM_COMPLEX_MODEL, LLM_PREFER_BILLION, LLM_TIMEOUT_MS. OLLAMA_HOST / OLLAMA_MODEL / OLLAMA_API_KEY alias the matching LLM_* names.
Other	EXECUTION_ENV, DB_*, BROWSERSTACK_*, LOG_LEVEL

Paths of note: reports/downloads/ (saved files), reports/self-heal/healed-locators.json (heal cache), recordings/test-runs/ (videos).

Run identity: RUN_ID env or file-locked reports/.run-id so all workers share one run.

Defaults: LLM_HOST=http://localaiserver.testingprofessor.net, LLM_API=ollama, LLM_MODEL=auto, LLM_ROUTER_MODEL=llama3, LLM_COMPLEX_MODEL=gemma4:latest, LLM_PREFER_BILLION=32, SELF_HEAL_ENABLED=false, CLICK_PAUSE_MS=800.

3. CustomWorld

```
browser, context, page, actions, logger, shared
runId, workerId, scenarioId, scenarioName, featureName, tags
stepIndex, screenshotPath, videoPath, tracePath, lastError, browserName
get waits / asserts          → from PlaywrightActions
get login / customer / application → business facades
```

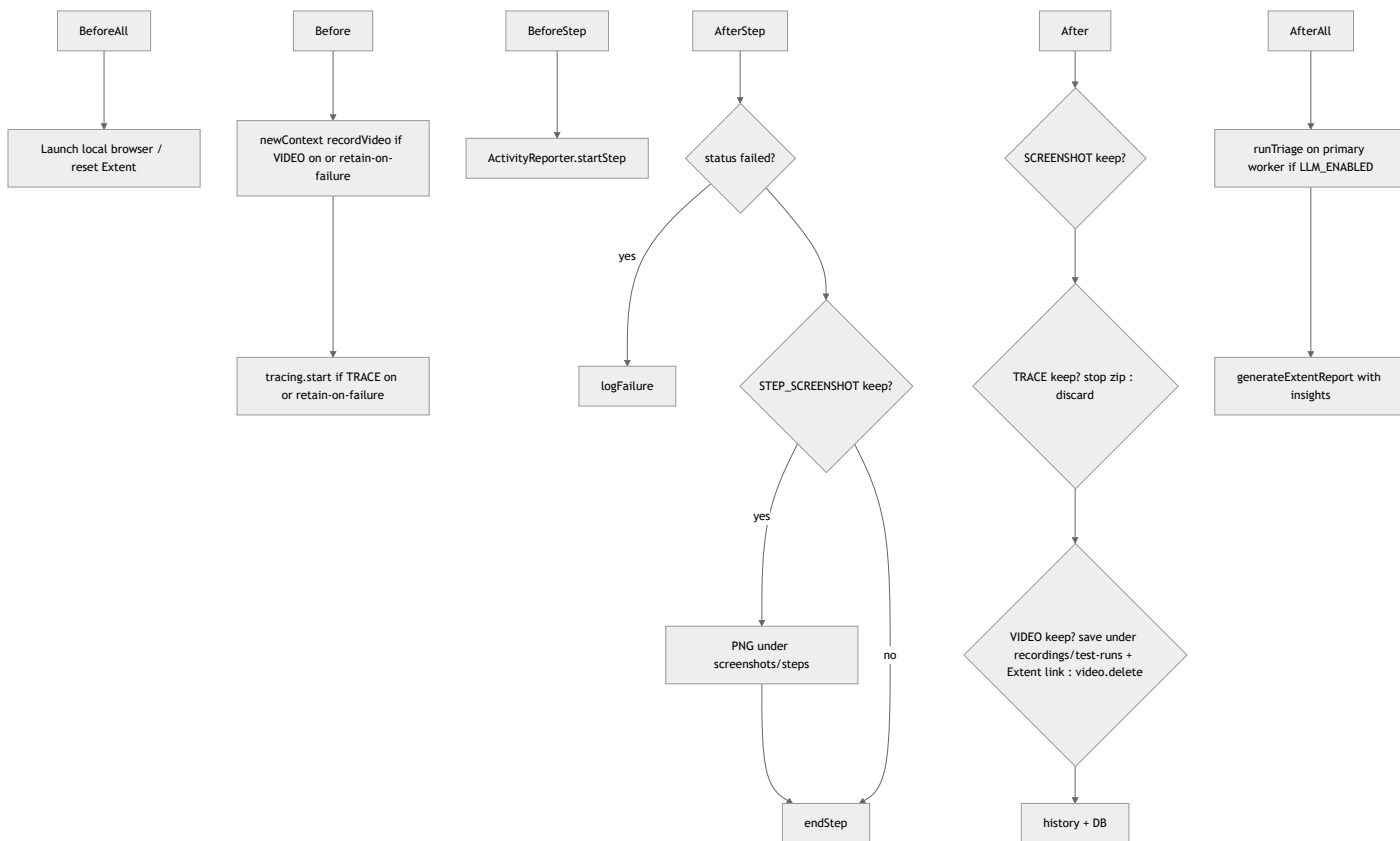
Constructed by Cucumber. Logger is rebound in Before with correlation metadata. stepIndex increments in BeforeStep for step-screenshot file names. Thin steps call this.customer.* / this.login.* / this.application.*.

3.1 Business layer

Class	Responsibility
BaseBusiness	Access to world, actions, logger, shared buffer
LoginBusiness	Launch app, staff sign-in, bank-manager entry
CustomerBusiness	Add customer, search list, open accounts, customer login/deposit
ApplicationBusiness	Cross-cutting app navigation helpers

Call chain: Feature → Step → Business → Page → PlaywrightActions.

4. Hooks



Hook	Behavior
BeforeAll	Create run, copy Allure categories, reset Extent data, launch local browser (not BrowserStack)
Before	New context/page, optional video dir and tracing, init SharedBuffer scenario scope, start Extent scenario
BeforeStep	Increment <code>stepIndex</code> , start Extent step, log STEP
AfterStep	Log failure on the step; capture step screenshot when <code>STEP_SCREENSHOT</code> applies; end Extent step
After	Scenario screenshot, trace zip or discard, video keep or delete; map error; persist JSON+DB; flush Extent
AfterAll	Close worker browser; finalize run; primary worker runs LLM triage (or custom report); generate Extent HTML; close DB

Artifact helpers: `src/config/artifacts.ts` — `shouldRecordArtifact` / `shouldKeepArtifact`.

5. PlaywrightActions API

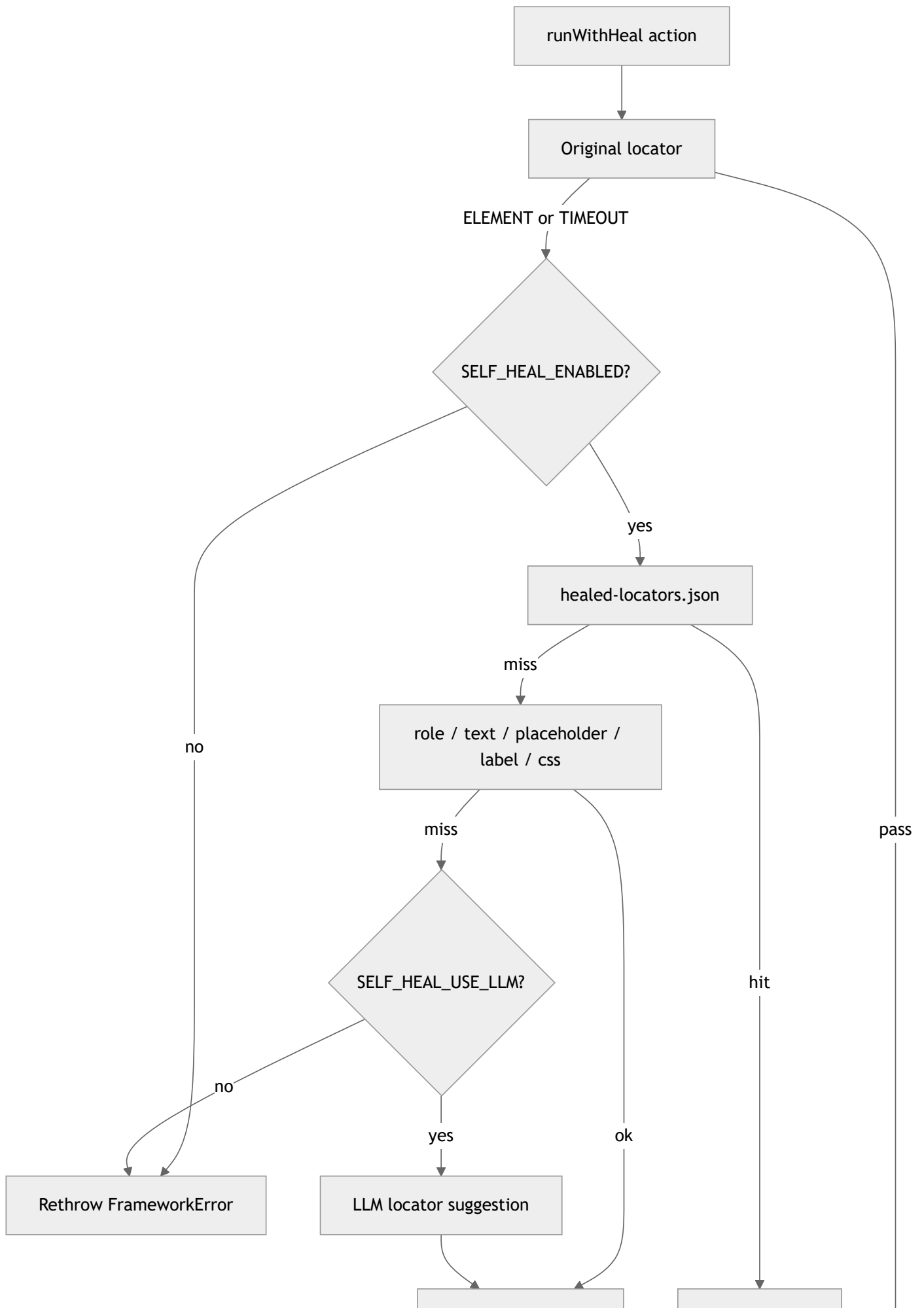
All methods log at debug/info, emit Extent activities, and wrap failures with `mapPlaywrightError`.
Locator verbs use `runWithHeal` when `SELF_HEAL_ENABLED=true`.

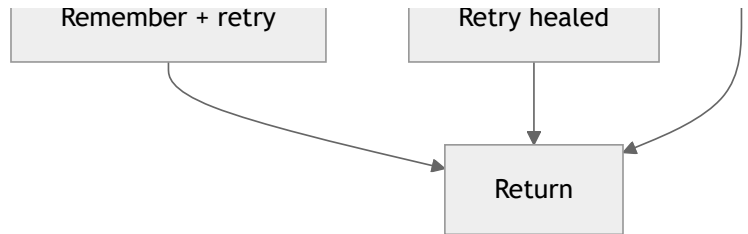
Group	Methods
Navigation / core	<code>goto</code> , <code>click</code> , <code>dblClick</code> , <code>fill</code> , <code>type</code> , <code>clear</code> , <code>select</code> , <code>selectRandom</code> , <code>hover</code> , <code>check</code> , <code>unchecked</code> , <code>press</code> , <code>pressOn</code>
Drag and drop	<code>dragAndDrop</code> , <code>dragAndDropByMouse</code> , <code>dragByOffset</code>
Frames	<code>frame</code> , <code>nestedFrame</code> , <code>inFrame</code> , <code>getFrameByUrl</code> , <code>getFrameByName</code> , <code>requireFrameByUrl</code> , <code>requireFrameByName</code> , <code>clickInFrame</code> , <code>fillInFrame</code>
Shadow DOM	<code>inShadow</code> , <code>shadowLocator</code> , <code>deepShadowLocator</code> , <code>clickInShadow</code> , <code>fillInShadow</code> , <code>dblClickInShadow</code>
Upload / download	<code>upload</code> , <code>uploadPayload</code> , <code>clearUpload</code> , <code>download</code> , <code>downloadByClick</code> , <code>downloadAndSave</code> , <code>downloadByClickAndSave</code>
Scroll	<code>scrollIntoView</code> , <code>scrollBy</code> , <code>scrollTo</code> , <code>scrollToTop</code> , <code>scrollToBottom</code> , <code>scrollElementBy</code> , <code>scrollElementTo</code> , <code>mouseWheel</code> , <code>getScrollPosition</code>
JavaScript	<code>evaluate</code> , <code>evaluateScript</code> , <code>evaluateOnLocator</code> , <code>evaluateHandle</code> , <code>dispatchDomEvent</code> , <code>setValueByJs</code> , <code>getPageProperty</code> , <code>getDocumentReadyState</code>
Wait / assert / read	<code>waitForVisible</code> , <code>waitForHidden</code> , <code>waitForUrl</code> , <code>waitForLoadState</code> , <code>getText</code> , <code>getAttribute</code> , <code>getCount</code> , <code>isVisible</code> , <code>isEnabled</code> , <code>assertVisible</code> , <code>assertText</code> , <code>assertTitle*</code> , <code>screenshot</code>
Windows	<code>launchApplication*</code> , <code>clickAndSwitchToNewWindow*</code> , <code>switchToWindow*</code> , <code>switchToParentWindow</code> , <code>closeCurrentWindow</code> , <code>waitForPopupCloseAndSwitchToParent</code> , <code>waitForWindowWithUrl</code>

`select` accepts a value, `{ label }`, `{ index }`, or blank (random option). `selectRandom` can exclude labels such as `No Category`. Downloads save under `reports/downloads/`.

Related utilities: `waitConditions` (including `observe` pause after clicks), `Assertions`, `SelfHeal.resolveHealedLocator`.

5.1 Self-heal





Successful heals are logged as Extent `selfHeal` activities. Prefer fixing page-object locators; treat heal as a resilience layer, not the primary design.

6. Page Object pattern

Core banking pages (not only templates):

Class	Role
LoginPage	Staff sign-in
AddCustomerPage	Create customer from Faker data
CustomersPage	Search / assert customer in list
OpenAccountPage	Open N accounts for a customer
CustomerDashboardPage	Customer login, Continue as first name, deposit

```

export class LoginPage extends BasePage {
  private readonly user = this.page.locator('#user');
  async login(user: string, pass: string) {
    await this.actions.fill(this.user, 'username', user);
  }
}

```

Feature: `features/customer.feature`. Steps: `features/steps/customer.steps.ts`. Export pages from `src/pages/index.ts`.

7. Error catalog

Class	Code	Category	Typical Playwright signal
ElementNotFoundError	PW_ELEMENT_NOT_FOUND	ELEMENT	Timeout waiting for locator
StrictModeViolationError	PW_STRICT_MODE	ELEMENT	strict mode violation
TimeoutExceededError	PW_TIMEOUT	TIMEOUT	TimeoutError

NavigationError	PW_NAVIGATION	NAVIGATION	net::, NS_ERROR, navigation
TargetClosedError	PW_TARGET_CLOSED	TARGET_CLOSED	Target closed
AssertionFailedError	PW_ASSERTION	ASSERTION	assert* helpers
BrowserLaunchError	PW_BROWSER_LAUNCH	BROWSER	launch / connect failure
ConfigurationError	FW_CONFIGURATION	CONFIGURATION	missing env / bad browser
DatabaseError	FW_DATABASE	DATABASE	SQL / connection
SharedStoreError	FW_SHARED_STORE	SHARED_STORE	file lock
AIServiceError	FW_AI_SERVICE	AI	LLM HTTP / missing API key / timeout
FrameworkError	PW_UNKNOWN	UNKNOWN	fallback

Mapper: `src/errors/mapPlaywrightError.ts`. JSON shape includes code, category, locator, url, action, timestamp, stack, cause.

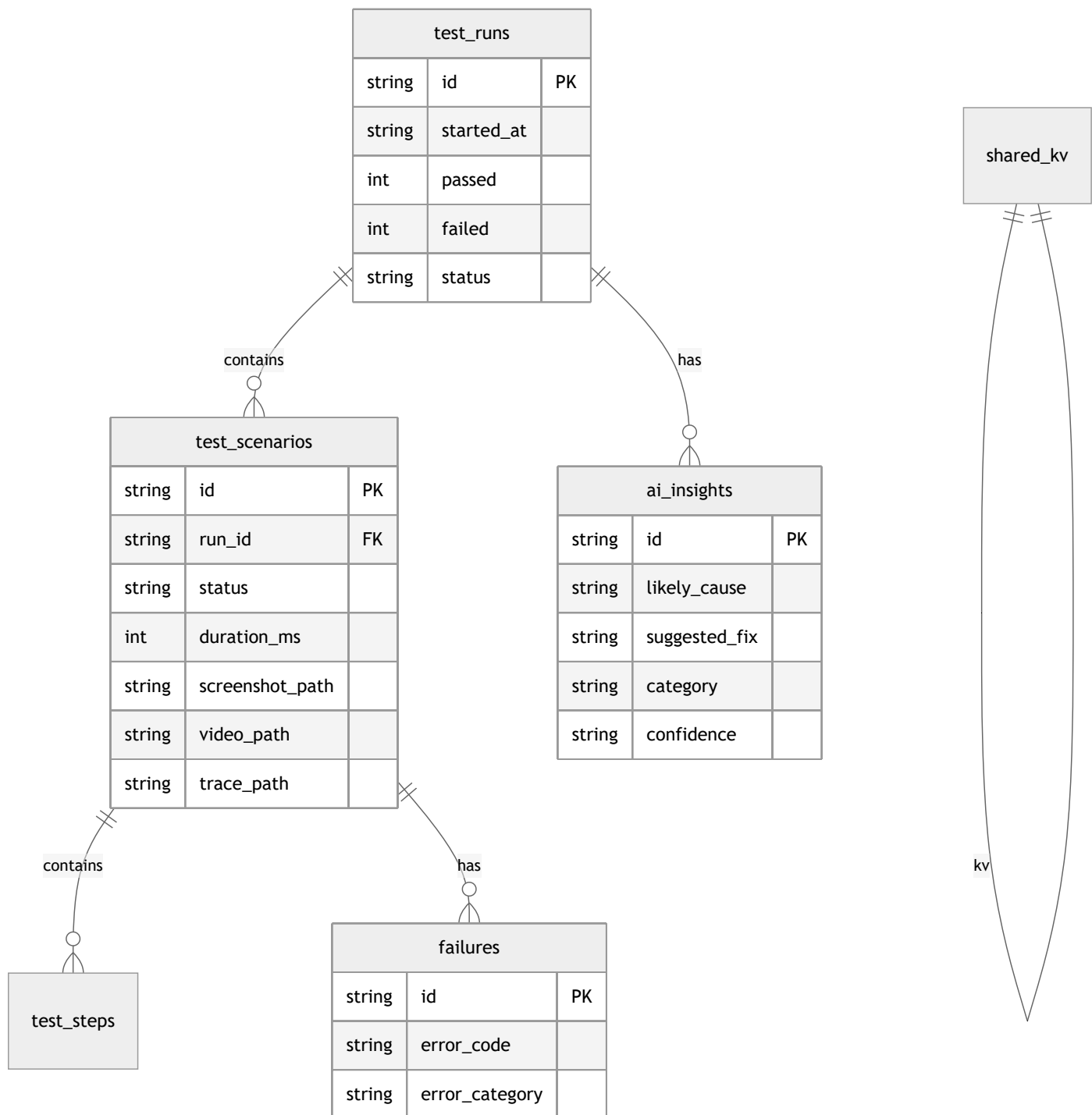
8. SharedBuffer

```
await world.shared.set('orderId', '123', { scope: 'global', ttlMs: 600000 });
const id = await world.shared.get('orderId', 'global');
```

Global writes: `reports/.shared-buffer.json` with `proper-lockfile`, then optional DB upsert into `shared_kv`.

Customer keys: `customer`, `customerName`, `customerFirstName`, `customerLastName`, `customers`, `lastCustomer`, `accounts`, `accountCount`.

9. Database ER



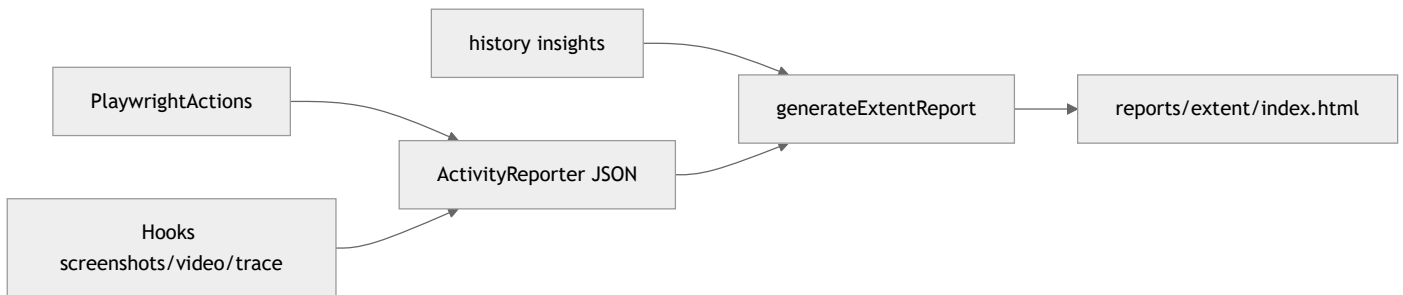
Placeholders use `?`; the Postgres client rewrites them to `$1..$n`. Upsert for `shared_kv` uses `REPLACE INTO` (MySQL) or `ON CONFLICT` (Postgres).

If `DB_ENABLED=false`, repository methods no-op and the dashboard reads `reports/history/*.json`. History insights may include `model` (chosen LLM) even if the DB column is not present.

10. Reporting modules

- `historyStore.ts` — append scenario, insights, maintain runs index.

- `customFailureReport.ts` — HTML with mapped reason, locator, screenshot, AI note + model name.
- `src/reports/extent/` — ActivityReporter (Gherkin steps + Playwright actions + step screenshots), `generateExtentReport.ts` writes `reports/extent/index.html` with video/trace links and AI box.
- Allure formatter in `cucumber.js`; categories copied from `src/reports/allure-categories.json`.



11. Dashboard APIs

Method	Path	Description
GET	/api/health	DB flag, LLM enabled/reachable, host, model
GET	/api/runs	Run list
GET	/api/runs/:id	Run + scenarios
GET	/api/failures	Optional ?runId=
GET	/api/trends	Last 30 runs for charts
GET	/api/insights	AI insights
POST	/api/triage	Trigger LLM triage for a run

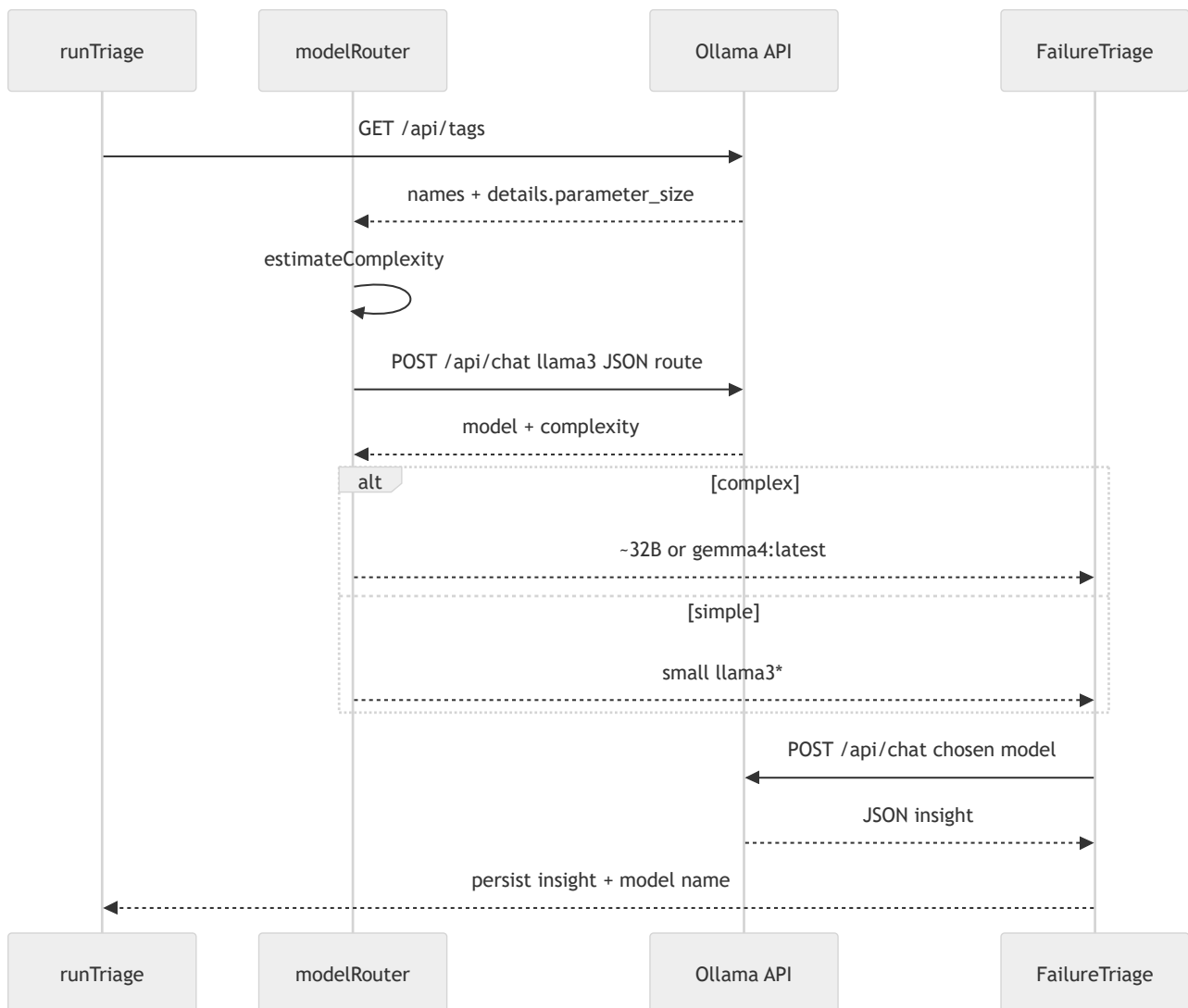
UI: overview KPIs, pass/fail trend (Chart.js), run table, failure drill-down, flaky detection, AI Insights tab showing category, confidence, and model name.

12. LLM triage and routing

Modules:

- `llmClient.ts` — Ollama `/api/tags` and `/api/chat` (OpenAI `/v1/*` fallback), Bearer + X-API-Key, timeout, catalog with `parameterBillion`.

- `modelRouter.ts` — estimate complexity; list catalog; ask small llama3 to pick; heuristic if router missing.
- `FailureTriage.ts` — JSON prompt for likelyCause / category / suggestedFix / confidence; stores `model`.
- `runTriage.ts` — iterate history failures, skip already-triaged, write custom report.
- `llmHealth.ts` — `npm run ai:health` prints installed models and 32B-class entries.



32B-class band is 27B–40B so names like `qwen2.5:32b`, `llama3.1:32b`, and `gemma2:27b` qualify. Size is taken from `details.parameter_size` or parsed from the tag (`:32b`, `8x7b MoE`). The llama3 router itself is constrained to models $\leq 8B$ so a 32B llama3 is not used for routing.

Heuristic when llama3 is not installed: complex $\rightarrow$ closest $\sim 32B$ then `gemma4:latest`; simple $\rightarrow$ smallest llama3*; medium $\rightarrow \sim 8B$.

Auth: nginx in front of the server returns 401 without `LLM_API_KEY`. Chat is skipped with a warning; reports still generate.

13. BrowserStack connect

```
wss://cdp.browserstack.com/playwright?caps=<urlencoded JSON>
```

Caps include browser, os, build, username, accessKey, scenario name. Local launch path is skipped when `EXECUTION_ENV=browserstack`; each scenario connects and closes its own remote browser.

14. CI/CD

- GitHub Actions: matrix chromium/firefox/webkit against `https://corebanking.testingprofessor.net/`; required secrets `APP_USERNAME` / `APP_PASSWORD` (optional `APP_ENCRYPTION_KEY`); `LLM_ENABLED=false`; tags `@smoke` and not `@example`; artifact upload; optional BrowserStack job.
- Jenkins: install, optional migrate, test, Allure + HTML publisher.
- Docker: official Playwright image; compose brings MySQL, Postgres, Ollama, dashboard, tests profile. Hosted LLM is selected with `LLM_HOST` / `LLM_API_KEY`.

15. npm scripts (execution styles)

Script	Use
test / test:chrome / firefox / webkit	Browsers
test:headed / test:slow / test:debug / test:keep-open	Local debug
test:parallel	Cucumber workers
test:login / test:customer / test:e2e / test:customer:datadriven	Tag filters
test:browserstack / test:trace	Remote / tracing
report:custom / report:extent / report:all	HTML reports + optional triage
ai:health / ai:triage	List models / run failure triage
dashboard:start / db:migrate	Ops
docker:up / docker:test	Containers
docs:pdf	HLD/LLD PDF via Playwright

Codegen URL: `npm run codegen:url --url=https://corebanking.testingprofessor.net --browser=firefox`

Tags: `npm run test:tags -- @smoke` and not `@wip`

16. Logging format

Console: timestamp, level, `runId|workerId|scenarioId`, message. Files: JSON in `logs/framework.log` and `logs/errors.log` with rotation. Step screenshots, self-heal retries, and LLM route decisions are logged at info.

17. Related documents

See [High Level Design \(HLD\)](#) for system context and deployment views. Stakeholder briefing: `docs/hld-stakeholder.docx` / `docs/hld-stakeholder.pptx` (`npm run docs:hld:stakeholder`). Interactive module summaries: Cursor canvases `hld-architecture` and `l1d-modules`.